



MALLA REDDY TECHNICAL CAMPUS
A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH
Deemed to be University
Approved by UGC & AICTE New Delhi
Maisammaguda, Medchal (Dist), Hyderabad - 500100, Telangana



COURSE STRUCTURE & SYLLABUS

**B.Tech – CSE(AI & ML), CSE(AI), CSE(AI & DS), CSE(CS), CSE(IOT & CS iNC. BCT),
ECE, R & AI, CSIT, CS & ME, CS & BS, CS & BiO sci.**

Admitted from A.Y. 2025-26 Batch

I Year I Semester							
3 Week Induction Programme							
S. No	Subject Code	Course Code	Course Title	L	T	P	C
1	25MCS101	BS	Mathematics for Computing	3	0	0	3
2	25FQM101	BS	Fundamentals of Quantum Computing	2	0	0	2
3	25LPS101	HS	Professional Skills for Engineers	3	0	0	3
4	25CPS101	ES	Computer Programming	4	0	2	5
5	25WAD101	PC	Web Application Development-1	2	1	4	5
6	25LSE101	BS	Quantitative Skills	2	1	0	3
7	25IKS101	MC	Indian Knowledge System	0	0	2	0
8	25FNL101	AEC	Foreign Language-I (Not limited to Spanish/ German/French/Korean/Mandarin etc.)	0	0	1	0
Total				16	2	9	21

MATHEMATICS FOR COMPUTING

25MCS101

L	T	P	C
3	0	0	3

I B.TECH I SEMESTER

Course Outcomes:

- CO1:** Gain proficiency in understanding and working with various number systems, including binary, decimal, octal, and hexadecimal.
- CO2:** Master logical and bitwise operators for mathematical and computational operations.
- CO3:** Apply arithmetic and modular mathematics concepts to solve computational problems and basic error detection.
- CO4:** Understand combinatorics concepts such as permutations and combinations, and apply probability rules in real-world scenarios.
- CO5:** Perform matrix operations such as addition and multiplication and explore their applications in graphs and transformations.
- CO6:** Apply set theory concepts and binary operations for data organization and problem- solving in computer science.

MODULE - I

NUMBER SYSTEMS AND OPERATIONS: Number Systems: Binary, Decimal, Octal, Hexadecimal. Conversion between number systems. Representation of signed and unsigned integers. Floating point representation. Logical and Bitwise Operators: AND, OR, NOT, XOR, NAND, NOR, Shift operations

MODULE - II

ARITHMETIC AND MODULAR MATHEMATICS

Basic arithmetic operations in different number systems. GCD and LCM: concepts and calculation methods. Modulo Arithmetic: basic operations and principles. Introduction to applications in computing. Simple error detection techniques.

MODULE - III

COMBINATORICS AND PROBABILITY

Basic Principles of Counting. Permutations: Arrangements with and without repetition, applications in real-world scenarios. Combinations: Selection problems, techniques and real-life examples. Probability Fundamentals: Basic concepts, probability rules, simple and conditional probability.

MODULE – IV

MATRICES AND SERIES

Matrix Representation and Types. Matrix Addition and Multiplication. Properties and Operations. Determinants. Matrix Applications: Graph Representation, Linear Transformations. Arithmetic Series. Geometric Series. Summation Techniques.

MODULE - V

SETS AND BINARY OPERATIONS

Fundamentals of Set Theory. Set Operations: Union, Intersection, Difference. Venn Diagrams and Applications. Cartesian Products. Binary Operations. Applications of Set Theory in Data Organization and Analysis.

REFERENCE BOOKS:

1. Discrete Mathematics and Its Applications" by Kenneth H. Rose
2. Algorithms and Complexity Theory

TEXT BOOKS:

1. "Algorithm Design Manual" by Steven S. Skiena
2. "Mathematical Logic" by Herbert B. Enderton

FUNDAMENTALS OF QUANTUM COMPUTING

25FQM101

L	T	P	C
2	0	0	2

I B.TECH I SEMESTER

Course Outcomes:

CO1: To be exposed to the fundamental concepts of Wave nature of Particles and Particle nature of Waves.

CO2: Understand the formation of energy bands in solids

CO3: Apply Quantum mechanics in the emerging field of Quantum computing.

CO4: Students will be able to interpret and construct basic quantum circuit diagrams.

CO5: Students will be able to identify and discuss real-world applications of quantum computing

MODULE - I

QUANTUM MECHANICS: Introduction to quantum physics, blackbody radiation, photoelectric effect-Broglie hypothesis, Davisson and Germer experiment, Heisenberg uncertainty principle, born interpretation of the wave function, time dependent and independent Schrodinger wave equation - particle in one dimensional potential box.

MODULE - II

BAND THEORY OF SOLIDS

Classical and Quantum Free electron theory assumptions, Merits and drawbacks (qualitative), Bloch theorem, Kronig- Penny model (qualitative), E-k diagram, Origin of energy bands in solids: Classification of Solids - Conductors, Semiconductors and Insulators.

MODULE - III

INTRODUCTION QUANTUM COMPUTING

Introduction, Classical & quantum bits, Superposition and Entanglement, Qubits Operators and Measurement: Bra-Ket (Dirac) notation, Bloch sphere representation of a Qubit, Classical logic gates, Quantum logic gates: Single qubit gates (Pauli-X, Pauli-Y, Pauli-Z, Hadamard), multiple qubit gates (CNOT gate)

MODULE – IV

QUANTUM CIRCUITS

Introduction to Quantum Circuits, Differences between classical and quantum circuits, Basic Circuit Notation, Quantum circuit diagrams, superposition, and entanglement using quantum circuits.

MODULE - V

BRIEF APPLICATIONS OF QUANTUM COMPUTING

Molecular modelling and design of new materials, deep learning, cyber security, quantum neural networks and artificial intelligence. Ethical considerations in quantum computing.

REFERENCE BOOKS:

1. Quantum Computing for Computer Scientists by Noson S. Yanofsky and Mirco A. Mannucci
2. Chris Bernhardt, Quantum Computing for Everyone, MIT Press, Illustrated Edition, April 2019.
3. Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II
4. Basic Tools and Special Topics, World Scientific. Pittenger A. O., an Introduction to Quantum Computing Algorithms
5. David McMahon, Quantum computing explained, Wiley-interscience, John Wiley & Sons, Inc. Publication 2008.
6. Quantum Computing in Practice with Qiskit and IBM Quantum Experience, Hassi Norlén, 2020.

TEXT BOOKS:

1. Michael A. Nielsen and Isaac L. Chuang, Quantum computation and quantum information, Cambridge University Press, 2010.

PROFESSIONAL SKILLS FOR ENGINEERS

25LPS101

L	T	P	C
3	0	0	3

I B.TECH I SEMESTER

Course Outcomes:

- CO1:** Understand the dynamics of English grammar, including nouns, pronouns, verbs, and modifiers, in functional English usage.
- CO2:** Develop the ability to comprehend and respond in basic English across various contexts.
- CO3:** Construct simple conversational sentences using conjunctions, prepositions, and diverse verb forms to communicate clearly and effectively.
- CO4:** Engage in numerous practical spoken English exercises to reduce public speaking anxiety, build confidence, and enhance presentability in verbal communication.
- CO5:** Improve listening skills and comprehension through exposure to various accents, speech.

MODULE I:

THE FUNDAMENTALS OF NOUNS AND PRONOUNS

Exploring Nouns vs. Main Verbs, Understanding Homophones and Homonyms, The Concept of Noun Number and Collective Nouns, Analyzing Pronouns: Referring to People and Objects, The Function of Possessive Nouns: Apostrophes and "Of" Forms, Usage of Possessive Pronouns

MODULE II:

MODIFIERS IN LANGUAGE

Adjectives: Identification, Usage, and Degrees, Adverbs – Types and Usage, Placement of Adverbs in Sentences, Articles – A/An and The, Zero Article and Its Usage, The Role of Articles in Specific and General References, Und

MODULE III:

EXPLORING VERBS

Understanding Verbs and Their Forms, Different Types of Verbs, Regular vs. Irregular Verbs, The Agreement of Subject and Verb

MODULE IV:

ELEMENTS FOR COMPLEX SENTENCES

Prepositions – Expressing Relationships, Conjunctions – Connecting Ideas in Sentences, Question Words – Basics of Formulating Questions, Contractions – Understanding Common Shortened Forms, and Interjections – Expressing Emotions and Reactions

MODULE V:

INTRODUCTION TO SENTENCE CONSTRUCTION

Differentiation of Time Periods, Differentiation Between State and Process, How to Describe Possessions in Different Time Periods, Describing General Routines and Daily Practices, Factual Description vs. Opinion, Current Status of Happenings, Commands, Comments, and Suggestions, Simple Present Scenario Identification, Simple Past Scenario Identification

TEXT BOOKS:

1. English Grammar in Use" by Raymond Murphy
2. Practical English Usage" by Michael Swan
3. The Elements of Style" by William Strunk Jr. and E.B. White

REFERENCE BOOKS:

1. Murphy, R. (2019). English grammar in use (5th ed.). Cambridge University Press

COMPUTER PROGRAMMING

25CPS101

L	T	P	C
4	0	2	5

I B.TECH I SEMESTER

Course Outcomes:

- CO1:** Understand the fundamentals of programming, including variables, data types, and operators, and be able to apply them in real-world scenarios.
- CO2:** Develop problem-solving skills using conditional statements, loops, and functions.
- CO3:** Gain proficiency in handling strings and working with data structures like lists, tuples, sets, and dictionaries.
- CO4:** Master object-oriented programming concepts such as encapsulation, inheritance, abstraction, and polymorphism, including method overloading and overriding.
- CO5:** Learn to debug effectively and handle errors and exceptions to create robust programs.
- CO6:** Create modular programs by working with modules and packages, ensuring code reusability and maintainability.

MODULE I:

INTRODUCTION

Definition: Software, Programming, Programming Language, Syntax, Why Python? First Program: Hello World, Variable & Data Types, Input Output Basics, Type Conversions, Operators: Arithmetic, Logical, Comparison, Assignment, Operator Precedence

MODULE II:

CONTROL STRUCTURES AND FUNCTIONS

Conditional Statements: If, Else, Else-If (Elif), Loops: For Loop, While Loop, Lists and Strings, String Manipulation: Case Conversion Methods, Counting and Searching Methods, String Formatting, Functions: Defining and Calling Functions, Parameters and Arguments, Return Statements, Local and Global Scope, Default Arguments, Keyword Arguments, Lambda Functions, Recursive Functions

MODULE III:

PYTHON DATA STRUCTURES

Lists: Creation, Accessing and Modifying Elements, Traversing Lists, Nested Lists, Copying Lists, Sorting and Merging Lists, Tuples: Definition and Usage, Operations on Tuples, Tuple Unpacking, Nested Tuples, Immutability, Sets: Creating Sets, Operations on Sets, Membership. Tests, Set Methods and Use-cases, Dictionaries: Creating and Updating Dictionaries, Accessing and Manipulating Key-Value Pairs, Nested Dictionaries, iterating through Dictionaries, List Comprehensions: Syntax, Applications, Creating Efficient and Readable List Operations

MODULE IV:

INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

Object Oriented Programming: Basics, Real-life Object Modeling, Organizing Descriptions, Classes, Defining Classes, Objects and Attributes: Instance Attributes, Class Attributes, Instance Methods, Class Methods, Static Methods, Encapsulation: Access Modifiers, Private Attributes, Getters and Setters, Modeling Classes, Inheritance: Understanding Inheritance, Implementing Class Hierarchies, Abstraction: Abstract Classes and Methods, Abstract Base Classes, Implementations, Polymorphism: Operator Overloading, Function Polymorphism, Method Overriding, Method Overloading.

MODULE V:

FILE HANDLING AND EXCEPTION HANDLING

File Handling: Reading, Writing, and Appending Files, Performing File System Operations, Error Handling in File Operations, Exception Handling: Understanding Exceptions, Types of Errors, Handling Exceptions with try-except blocks, Raising Custom Exceptions, Imports and Modules: Importing Libraries, Creating and Managing Modules, Packages: Creating, Managing, and Working with Packages, Using Third-Party Libraries

LIST OF EXPERIMENTS:

1. Given a two-digit number, print "Lucky Number" if any of the following conditions are satisfied: the number is a multiple of 9 or one of the digits is 9. Print "Unlucky Number" in all other cases.
2. Given 4 numbers, write a program to print the smallest number of the four numbers.
3. Print an Inverted Right-Angled Triangle of N rows using stars (*).
4. Print a Hollow Pyramid of N rows using stars (*).
5. Given a number N, write a program to print all the Prime Numbers from 1 to N.
6. Given a string S and N integers, where N is the length of the string S, print the string after shuffling the characters as per the order of the integers given.
7. Print Fibonacci Sequence
8. Implement Customer Details Class

REFERENCE BOOKS

1. Python Programming: A Modern Approach, Vamsi Kurama, Pearson
2. Python Programming A Modular Approach with Graphics, Database, Mobile, and Web Applications, Sheetal Taneja, Naveen Kumar, Pearson
3. Programming with Python, A User's Book, Michael Dawson, Cengage Learning, India Edition
4. Think Python, Allen Downey, Green Tea Press
5. Core Python Programming, W. Chun, Pearson
6. Introduction to Python, Kenneth A. Lambert, Cengage

TEXT BOOKS:

1. Supercharged Python: Take your code to the next level, Overland
2. Learning Python, Mark Lutz, O'reilly

WEB APPLICATION DEVELOPMENT -1

25WAD101

L	T	P	C
2	1	4	5

I B.TECH I SEMESTER

Course Outcomes:

- CO1:** Understand the role and importance of web and frontend development, including foundational syntax, debugging techniques, and building structured content with HTML and CSS.
- CO2:** Gain proficiency in creating responsive, mobile-friendly layouts using Bootstrap, Flexbox, media queries, and advanced CSS properties like Grid and animations.
- CO3:** Learn JavaScript basics, including variables, operators, conditionals, loops, and functions, to build interactive web pages.
- CO4:** Explore DOM manipulation techniques to dynamically update web content and styles, and handle user input using forms, events, and event listeners.
- CO5:** Use modern web technologies such as Fetch API and JSON to manage data and interact with web servers.
- CO6:** Design and implement responsive web applications that function seamlessly across different devices.

MODULE I:

HTML/CSS BASICS

Importance of Web/Frontend, Fundamentals: Syntax, Debugging, Tweaking Code, Getting Started with HTML and CSS, HTML Elements: Headings, Paragraphs, Buttons, Images, Containers, Lists, CSS Rulesets: Units, Padding, Margins, Box Model, Selectors: Class, ID, Type, Specificity, Block vs. Inline Elements, Inheritance, Cascading

MODULE II:

BOOTSTRAP AND FLEXBOX

Bootstrap Grid System, Container Components, Row Implementation, Column Structure, Display Properties, Flex Utilities, Sizing Utilities, Spacing Utilities, Color Systems, UI Components (Navbar, Buttons), Dynamic Elements (Carousel), Media Integration, Flexbox, Flex Container Properties, Flex-wrap Mechanics, Flex-direction Applications, Alignment Properties (Align-items, Align-self), Justify-content, Responsive Design Principles

MODULE III:

CSS MEDIA QUERIES, LAYOUT & ANIMATIONS

Media Query Fundamentals, Media Types Implementation, Media Feature Detection, Logical Operators in CSS, Layout Sizing Properties, Width and Height Manipulation, Minimum/Maximum Constraints, Overflow Control, CSS Grid System, Grid Layout Implementation, CSS Transition Properties, Transform Functions, Animation Keyframes, Animation Properties, Animation Timing

MODULE IV:

JAVASCRIPT BASICS AND DOM MANIPULATION

JavaScript Language Basics, Data Type Systems, Variable Declaration, Operators and Expressions, Control Structures, Conditional Statements, Loop Implementations, Function Declaration, Parameter Passing, Document Object Model, DOM Hierarchy, Element Selection Methods, Node Properties, Text Content Manipulation, Attribute Modification, Style Property Access

MODULE V:

EVENTS AND HTTP REQUESTS

Form Structure, Input Element Types, Form Control Components (Textarea, Select), Form Validation, Event Model, Event Listener Registration, Event Propagation, Event Object Properties, Asynchronous JavaScript, Callbacks, Scheduler Functions, HTTP Communication, Fetch API Implementation, JSON Data Structure, JSON Parsing and Serialization

LIST OF EXPERIMENTS:

1. Build Portfolio website using HTML, CSS and Bootstrap
2. Build a Todos Application Website by implementing the storage mechanisms
3. Build a Food munch website by using Bootstrap grid system
4. Build a simple Bookmark Maker using HTML Form Elements
5. Build a Speed typing test application with random quotation using fetch
6. Build a Library Management page to search the books using fetch
7. Build a Word cloud page using random function
8. Build a Wikipedia Search Application using fetch URL
9. Build a chatbot page by dynamically creating the elements
10. Build a Movie Reviews Page using HTML Form Elements

REFERENCE BOOKS:

- Front-End Web Development (UI/UX, HTML, CSS, JavaScript)
- "JavaScript and JQuery: Interactive Front-End Web Development" by Jon Duckett
- "Eloquent JavaScript" by Marijn Haverbeke

TEXT BOOKS:

- "Learning React: Functional Web Development with React and Redux" by Alex Banks and Eve Porcello
- "CSS Secrets: Better Solutions to Everyday Web Design Problems" by Lea Verou

QUANTITATIVE SKILLS

25LSE101

L	T	P	C
2	1	0	3

I B.TECH I SEMESTER

Course Outcomes:

- CO1:** Develop a strong foundation in quantitative aptitude concepts essential for competitive exams, recruitment tests, and higher education entrance exams.
- CO2:** Enhance problem-solving skills and analytical thinking required to tackle numerical and logical reasoning questions effectively.
- CO3:** Gain the ability to approach quantitative aptitude problems strategically and improve speed and accuracy in solving mathematical questions.
- CO4:** Understand the practical applications of quantitative aptitude in real-world scenarios, such as finance, business analytics, and technical problem-solving.
- CO5:** Strengthen mathematical concepts, including arithmetic, algebra, and percentages which are commonly tested in aptitude assessments.
- CO6:** Improve decision-making and time management skills by practicing different types of quantitative problems under time constraints.

MODULE – I: NUMBER SYSTEMS AND ITS APPLICATIONS

Number Systems:

Classification of numbers, divisibility rules, concept of co-primes, power cycles, remainder cycles, trailing zeros, and complex numerical problems. Applications: Understanding multiples, factors, least common multiple (LCM), and greatest common divisors (GCD) along with their properties and applications.

MODULE – II:

PERCENTAGES AND ITS APPLICATIONS:

Percentages: Percentage formula, conversion between decimals and fractions, percentage increase and decrease. Applications: Solving problems related to profit and loss, discounts,

MODULE III – Simple and Compound Interest

Simple interest, and compound interest in business and financial scenarios.

MODULE – IV:

RATIOS, PROPORTIONS, AND ITS APPLICATIONS:

Ratios & Proportions: Understanding ratio concepts, direct and inverse proportion, and their mathematical applications. Applications: Mixtures and allegations, partnership-based profit sharing, age-based problems, and calculating averages in different contexts.

MODULE – V:

TIME, SPEED & DISTANCE:

Time, Speed, and Distance: Understanding the relation between time, speed, and distance, unit conversions, average speed, relative speed, effective speed, and solving problems related to trains and races.

REFERENCE BOOKS:

1. Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal
2. Fast Track Objective Arithmetic" by Rajesh Verma
3. Quantitative Aptitude for Competitive Examinations" by Abhijit Guha

TEXT BOOKS:

- "Kiran's SSC Mathematics" by Kiran Prakashan
- Arithmetic for Competitive Examinations" by R.S. Aggarwal

INDIAN KNOWLEDGE SYSTEM

25IKS101

L	T	P	C
0	0	2	1

I B.TECH I SEMESTER

Course Objectives

- Understanding the fundamental principles of Indian health systems such as Ayurveda and yoga which are useful in maintaining the health of a healthy person
- Practical implementation of health principles to correct the intake of our food, air, water and sunlight to achieve perfect health.
- Understanding traditional way of cleansing the body regularly, strengthening body with Yogic exercises, maintaining the internal balance to prevent diseases.
- Understanding our unique Mind Body Constitution and choosing the right lifestyle suitable to maintain the internal balance.
- Understanding the influence of external environment on internal health and ways to synchronise our body and mind with nature to ensure smooth functioning of all organ systems of our body.
- Understanding mind and its dynamics through knowledge of Ayurveda and Yoga and using the knowledge to maintain harmony between body and mind to achieve perfect mental health.

MODULE – I:

Understanding human body

Introduction to Ayurveda, the Knowledge of Life, Health and treatment aspects in Ayurveda, Influence of Pancha maha bhuta on Internal environment of Human being, understanding composition of Human body through the concept of Dosha Dhatu Mala, Understanding Prakruthi, the Mind – Body Constitution

MODULE – II :

Understanding the communication between body & Mind

Establishing communication between body and mind by understanding the language of body. Understanding the concept of Agni, Koshta, Sara and Ojas and their relevance in enhancing our immunity to protect from various infections. Looking at the world through the lenses of Dravya, Guna and Karma Applying the principle of Samanya and Vishesha in every aspect of life to achieve perfect health.

MODULE – III :

Introduction to Health Regimen

Understanding Swastha vritta, the healthy regimen to maintain state of wellbeing Dinacharya, the Daily regimen including Daily detoxification, exercise, Intake of Food, Water, Air and Sunlight, work and ergonomics, Rest and sleep hygiene. Ritu charya, the seasonal regimen, Sadvritta and the concept of social wellbeing, understanding trividha upastambhas, three pillars to health, Concept of Shadrasa in choosing appropriate nourishment to the body and mind.

MODULE – IV:

Introduction to Yoga

Definition, Meaning and objectives of Yoga, Relevance of yoga in modern age. Brief Introduction of Hatha yoga, Raja yoga, Karma yoga, Gyana Yoga, Bhakti yoga Understanding eight steps of Ashtanga yoga, Understanding Shatkriyas , the six cleansing procedures of Yoga

MODULE-V:

Introduction to Indian Psychology

Concept of Manas in Ayurveda and understanding Mind Body harmony, Triguna based Psychology in Ayurveda and Yoga, Influence of Tri dosha on Mind, Mind body intellect and consciousness complex, Understanding Consciousness and solution to issues within Human Mind.

References:

1. The Charaka Samhita
2. Teh Ashtanga Hridaya
3. Dr Deepak Chopra, Perfect Health--Revised and Updated: The Complete Mind Body Guide, Harmony publication, 2001
4. Vasant lad, Ayurveda, the Science of Self-healing: A Practical Guide: Science of Self- healing, lotus press, 1984
5. The Hatha yoga pradipika
6. The Patanjali yoga sutras
7. The Gheranda samhita
8. Swamy Satyananda Saraswati, Asana, Pranayama, Mudra and Bandha, Bihar School of Yoga, 2002

FOREIGN LANGUAGE – I

25FNL101

L	T	P	C
0	0	1	0

I B.TECH I SEMESTER

COURSE OUTLINE:

German I is meant to be an introduction to the German language and a basic orientation towards Germany (and to some extent Austria and Switzerland). Learning to understand and articulate oneself in day-to-day real-life situations, and to begin to make sense of Germany as a cultural space are the overall objectives of the course. Serious learners should be able to grasp the basic sentence structure and build a good foundational vocabulary through this course.

COURSE PLAN:

- Week 1:** Themes: Introducing oneself and others; Grammar: W questions, personal pronouns, simple sentence, verb conjugation
- Week 2:** Themes: hobbies, the week, numbers, the alphabet, months, seasons / Grammar: articles, plural, the verbs to have and to be
- Week 3:** Theme: In the city / naming places and buildings, means of transport, Basic directions / Grammar: definite and indefinite articles; negation – kein and nicht; imperative
- Week 4:** Themes: food, drink, family / groceries and meals / Grammar: the accusative
- Week 5:** Theme: Everyday life, telling time, making appointments / Grammar: prepositions am, um, von..bis; modal verbs, possessive articles
- Week 6:** Leisure activity, celebrations / Grammar: separable verbs, the accusative, past tense of to have and to be
- Week 7:** Contacts, writing letters / Grammar: dative
- Week 8:** My apartment, rooms, furniture, colours / Grammar: changing prepositions

Week 9: Professions / Grammar: perfect tense

Week 10: Clothes / Grammar: perfect tense and dative

Week 11: Health and the body / Grammar: the imperative and modal verbs

Week 12: Holiday and weather